

Project Proposal

Data Analysis and Visualization

AnimeFusion

Team Members:

Sohaib Ahmed **21L-5635** Section : BDS-5A

Muhammad Hamza **21L-5636** Section : BDS-5A

Muhammad Hassan Khalid **21L-5692** Section : BDS-5A

Project Title: Exploring Anime Trends: A Comprehensive Analysis of MyAnimeList Data

Project Description: Our project aims to conduct a detailed analysis of anime data from MyAnimeList. MyAnimeList <https://myanimelist.net/> is a popular platform where users rate, review, and discuss anime. We plan to explore various aspects of the anime world, including user preferences, genre trends, and the impact of rankings on popularity.

Dataset Source: Our primary data source will be MyAnimeList, and we will perform web scraping to extract relevant information. The dataset will include details such as anime titles, ratings, genres, user reviews, rankings, and popularity metrics.

Data Scraping Tool: To scrape data from MyAnimeList, we will utilize the Selenium and BeautifulSoup libraries in Python. Selenium will be used for dynamic

web page interactions, allowing us to navigate through multiple pages, while BeautifulSoup will aid in parsing and extracting the required information.

API for Additional Data: In addition to web scraping, we plan to complement our dataset by utilizing the MyAnimeList API. The API will efficiently provide us with additional details about each anime, including information about studios, airing dates, synopses, and alternative titles. We will use this API to expedite the data collection process and gain a more comprehensive understanding of each anime.

Objectives:

- Collect a comprehensive dataset from MyAnimeList, including information on anime titles, ratings, genres, and user reviews.
- Analyze user preferences by examining the distribution of ratings and genres.
- Investigate the relationship between rankings and popularity metrics.
- Explore trends over time by considering the release dates of anime and changes in user preferences.
- Visualizations depicting user preferences, genre trends, and the impact of rankings on popularity.
- Insights into the dynamics of the anime community based on the analysis of MyAnimeList data.
- Develop a comprehensive anime recommendation system by integrating collaborative and content-based filtering techniques.
- Assess the effectiveness of the collaborative and content-based models.